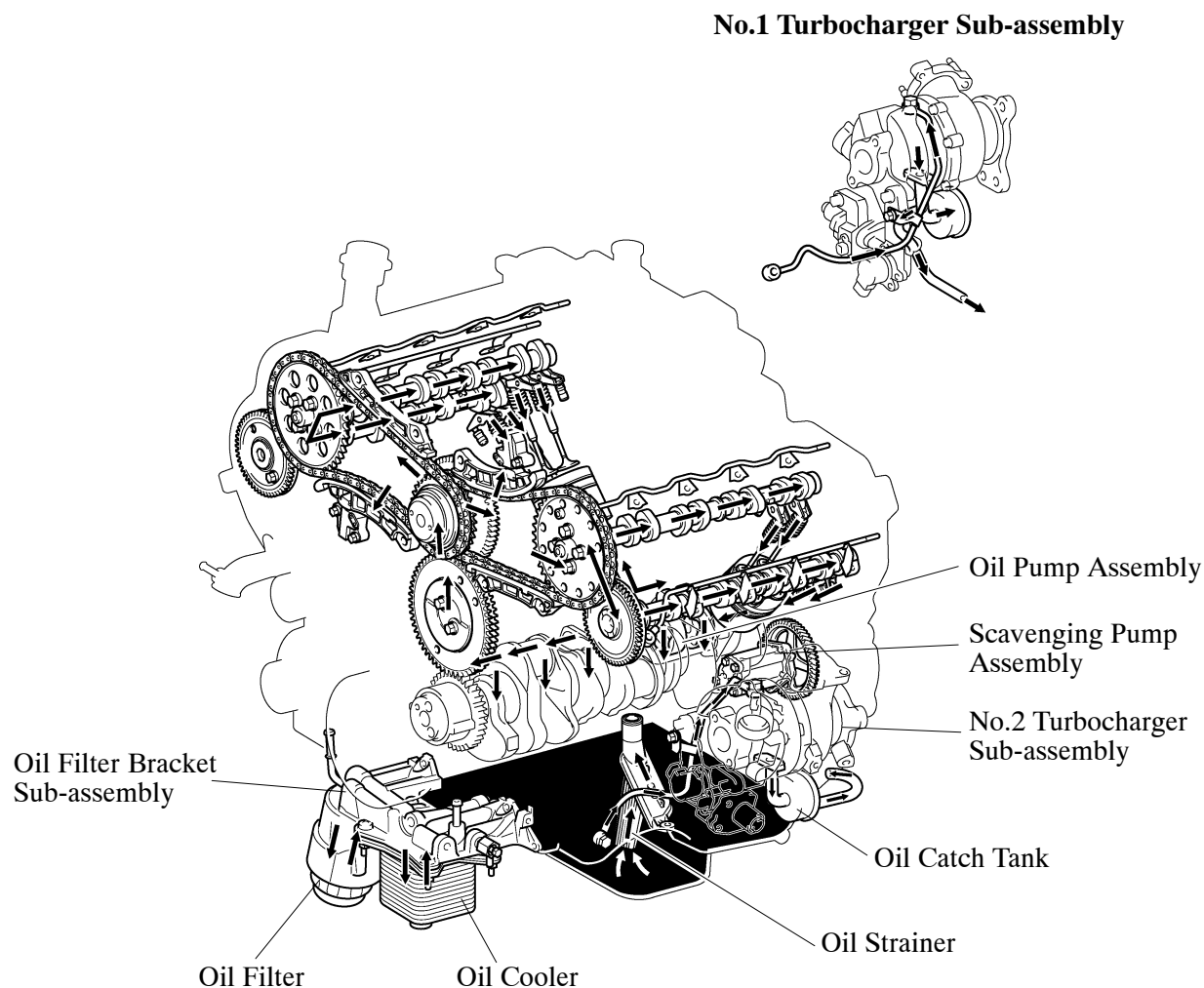


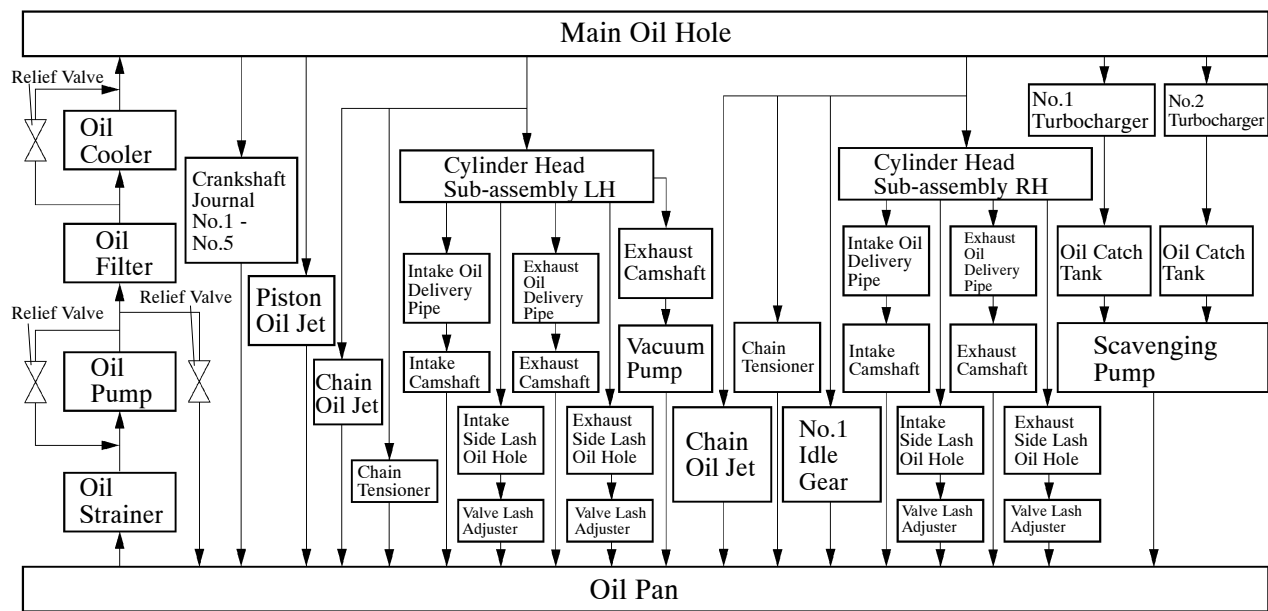
■ LUBRICATION SYSTEM

1. General

- The lubrication circuit is fully pressurized and all oil passes through an oil cooler and oil filter.
- A trochoid gear type oil pump, which uses 2 oil pump gears, is provided.
- A scavenging pump is used to suppress the accumulation of oil in the turbocharger while driving on a slope.
- A water-cooled type oil cooler is installed on oil filter bracket.



► Oil Circuit ◀



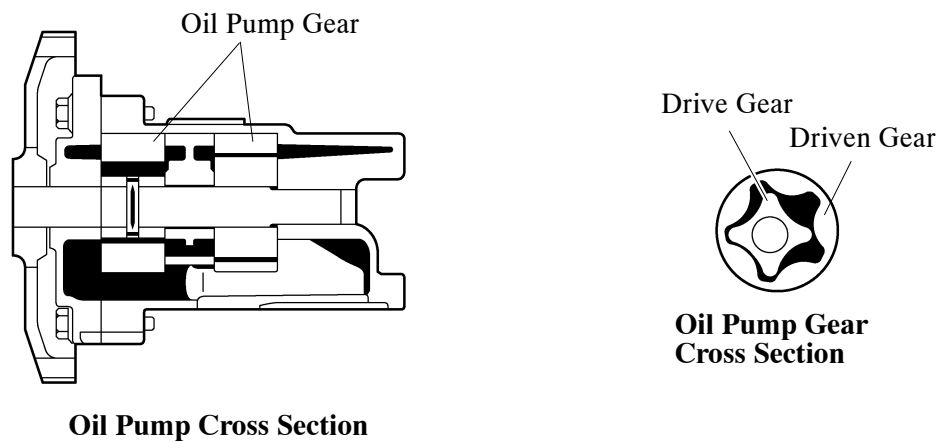
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► Oil Capacity ◀

Oil Capacity [Liters (US qts, Imp. qts)]	Dry	9.7 (10.3, 8.5)
	with Oil Filter	9.0 (9.5, 7.9)
	without Oil Filter	8.0 (8.5, 7.0)

2. Oil Pump

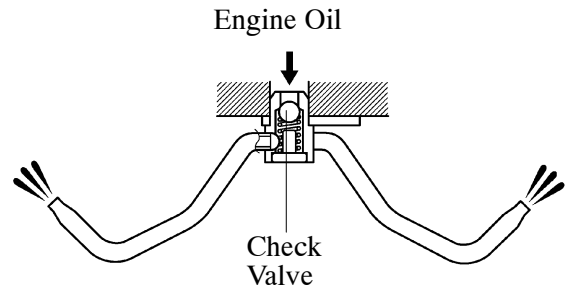
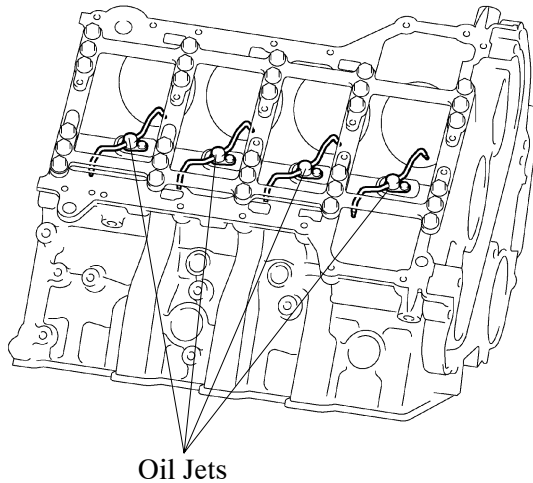
A trochoid gear type oil pump is used. The oil pump uses 2 oil pump gears and is driven by a gear engaged with the crankshaft.



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3. Oil Jet

- Oil jets are provided at the bottom of the cylinder block to spray oil to the piston's cooling channel, thus further cooling and lubricating the pistons.
- These oil jets contain a check valve to prevent oil from being fed when the oil pressure is low. This prevents the overall oil pressure in the engine from dropping.

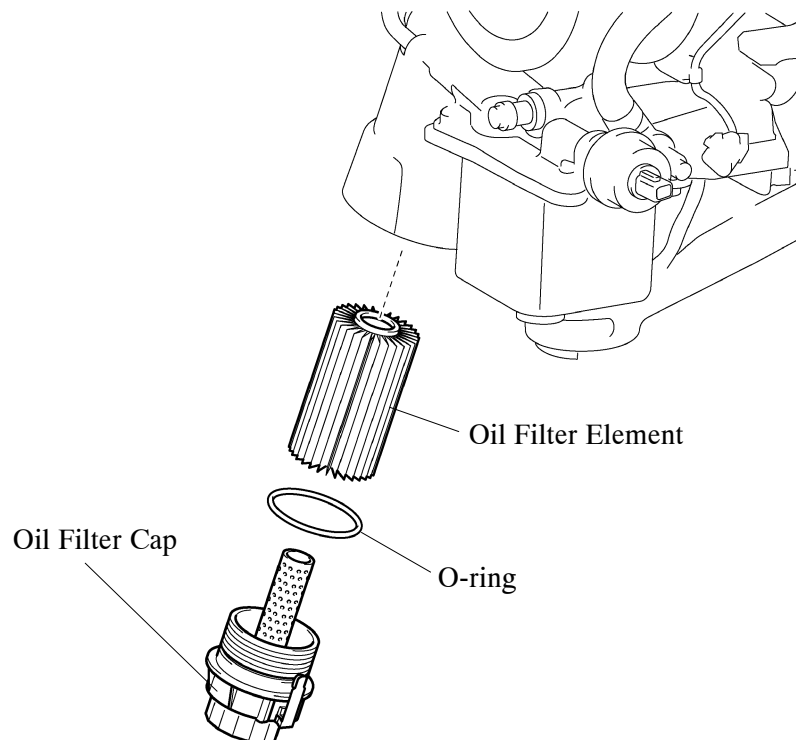


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4. Oil Filter

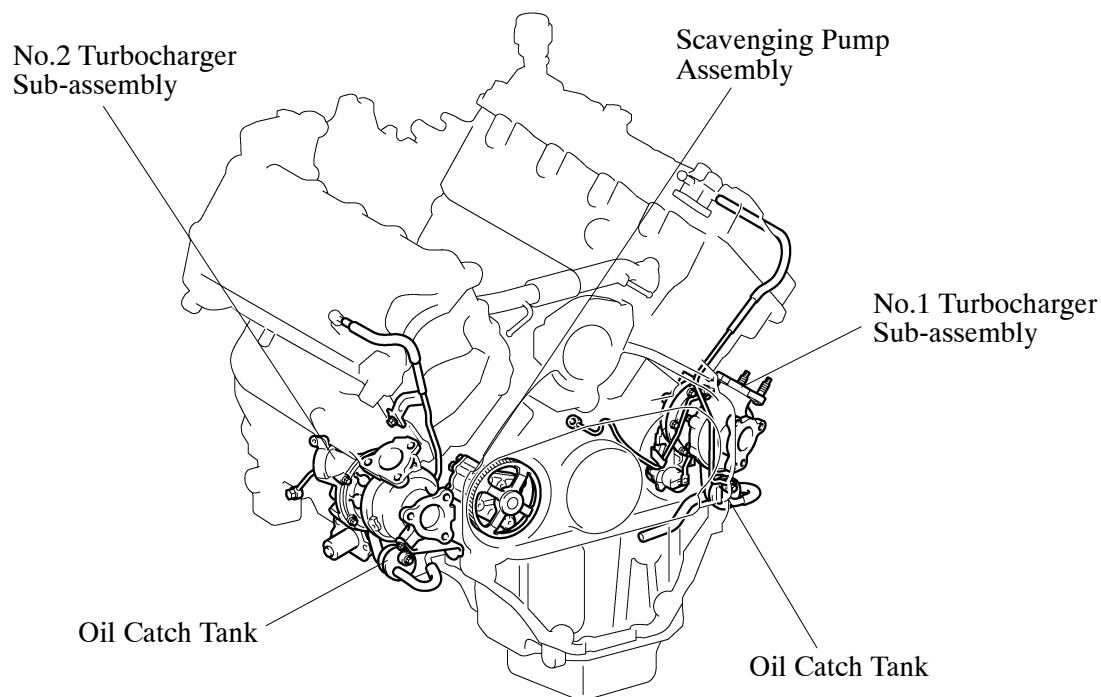
- An oil filter with a replaceable element is used. The oil filter element uses a high-performance filter paper to improve filtration performance. It is also combustible for environmental protection.
- A plastic oil filter cap is used.



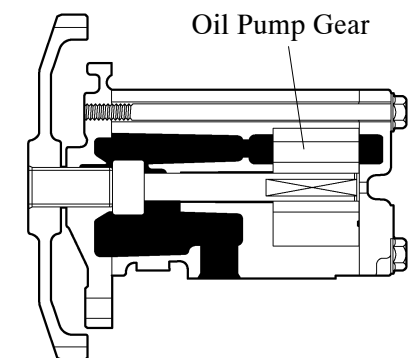
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5. Scavenging Pump

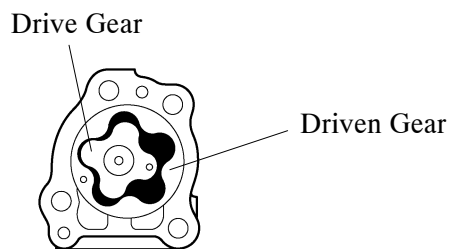
- The scavenging pump, which is located at the back end of the cylinder block, is driven by the crankshaft via gears.
- After the turbocharger discharges the oil, the scavenging pump returns the oil from the oil catch tank through a forced suction and discharges it to the oil pan.



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Scavenging Pump Cross Section



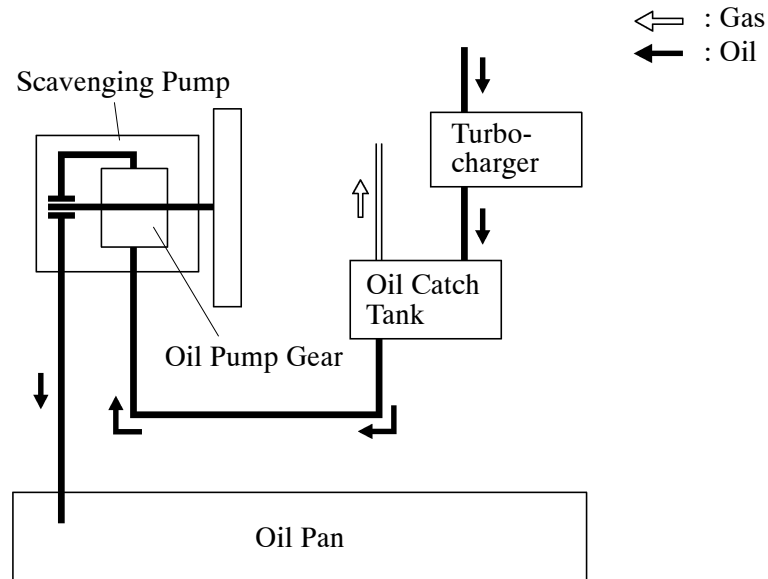
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Scavenging Pump Gear Cross Section

► Pump Components and Functions ◀

Component	Function
Oil Catch Tank	The oil catch tank separates the oil discharged by the turbocharger into gas and liquid.
Scavenging Pump	Returns the oil from the oil catch tank by suction, and discharges it to the oil pan.

► System Diagram ◀



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